

NyayaMitra AI

Team Collaboration Guide and Task Assignment Map

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1. Repository Overview & How to Clone

The skeleton project is pushed to the GitHub repository: `hanshikavelaga/Nyaya_mitra`. Every team member must clone the codebase to their local environments to begin their assigned tasks.

Clone Command

Open your terminal and run the following command to clone the code:

```
git clone https://github.com/hanshikavelaga/Nyaya_mitra.git
cd Nyaya_mitra
```

2. Feature Branch Git Workflow

To keep the codebase stable and prevent overwrite conflicts during development, all developers must follow a branch-based workflow. Do not merge code directly into main.

Daily Staging and Contribution Flow:

- **1. Sync Main:** Pull the latest updates from main before writing code.

```
git checkout main
git pull origin main
```

- **2. Branch Creation:** Create a local branch named after your feature.

```
git checkout -b feature/your-feature-name
```

- **3. Commit Changes:** Make commits locally with clear messages.

```
git add .
git commit -m "feat(backend): Setup RAG law query endpoint"
```

- **4. Push Branch:** Push your branch to GitHub.

```
git push -u origin feature/your-feature-name
```

- **5. Open Pull Request:** Go to GitHub, select your branch, and click 'Create Pull Request'. A peer must review the code before it is merged into main.

3. Development Cautions and Best Practices

[ATTENTION] Never Push Directly to Main

Pushed code directly to the main branch is strictly prohibited. Direct pushes bypass syntax checks and tests, leading to broken developer environments.

[ATTENTION] Never Commit Secrets / API Keys

Do not write or commit your Gemini API Key in the source files. Set it inside environment variables. The .gitignore is already set to ignore .env files.

Handling Merge Conflicts

- **Conflict Resolution:** If GitHub blocks a pull request due to merge conflicts, pull main into your branch locally, resolve conflicts in your code editor, and commit the changes:

```
git checkout feature/your-feature-name
git merge main
# Open files, resolve conflict blocks, then:
git add .
git commit -m "Merge main and resolve conflicts"
git push
```

CORS Setup Warning

- **CORS Policy:** The frontend communicates with the backend across different ports (Port 3000 -> 8000). The FastAPI CORS middleware in main.py must remain active. Do not modify CORS exclusions without consulting the Lead Developer.

4. Member Task Assignment Summaries

A comprehensive Excel file detailing files, days, and instructions is available at 'nyayamitra_project_plan_detailed.xlsx'. Below is the summary task checklist:

Member 1 - Backend & Database Developer

- **FastAPI Skeleton:** main.py, config.py, schemas.py (Day 1) - Highest Priority
- **Legal KB Creation:** data/laws_kb.json (Day 2) - High Priority
- **RAG Search Service:** services/rag_retriever.py (Day 2) - High Priority
- **Chat Q&A Setup:** services/chat_manager.py, routers/chat.py (Day 3) - Medium Priority
- **Calendar Event Exporter:** services/calendar_generator.py (Day 4) - Low Priority

Member 2 - Frontend UI Developer

- **UI Core Styling:** src/App.tsx, src/index.css (Day 1) - Highest Priority
- **File Uploader UI:** src/components/UploadZone.tsx (Day 2) - Highest Priority
- **Split Pane Shell:** src/components/WorkspaceLayout.tsx, DocViewer.tsx (Day 2) - High Priority
- **Timeline countdowns:** src/components/TimelineWidget.tsx (Day 3) - High Priority
- **Checklist & Q&A Chat:** src/components/ChecklistWidget.tsx, ChatPanel.tsx (Day 3) - Medium Priority

Member 3 - AI Pipeline Developer

- **Gemini Integration & OCR:** services/gemini_client.py, routers/upload.py (Day 1) - Highest Priority
- **Structured Output JSON:** services/analysis_synthesizer.py (Day 2) - High Priority
- **Chat Prompt Grounding:** services/chat_synthesizer.py (Day 3) - Medium Priority
- **Telugu Translation API:** services/translator.py (Day 3/4) - Medium Priority

Member 4 - QA, Accessibility & Presentation Lead

- **Mock Notices Setup:** data/sample_summons.pdf, sample_eviction_notice.png (Day 1) - Highest Priority
- **FastAPI Integration Tests:** tests/test_api.py (Day 2) - High Priority
- **Web Speech TTS Widget:** src/components/VoiceReader.tsx (Day 4) - High Priority
- **WhatsApp Reminder concept:** services/sms_scheduler_mock.py (Day 3/4) - Low Priority
- **IEEE Pitch preparation:** README.md, presentation_guide.md (Day 4) - High Priority

5. Key Integration Checklist

- **Ports:** Verify uvicorn backend on port 8000 and React Vite on port 3000 running in parallel.
- **Data Flow:** Verify uploader triggers Gemini OCR and fills RAG law citation matches dynamically.
- **Accessibility:** Verify Web Speech Synthesis plays audio summaries correctly in Telugu voice configs.